Mathematical Objective Formulation

The total loss function balances reconstruction accuracy and adversarial regularization:

$$L_{total} = alpha * L_{rec} + beta * L_{adv} \tag{1}$$

where alpha and beta denote hyperparameter weights chosen empirically.

$$grad_{theta} = nabla_{theta} * E[log D(x)] \tag{2.1}$$

Training converges steadily when the learning rate is scheduled adaptively.